Multi-Site Brain MRI Landmark Detection

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Abstract

We build and evaluate a six-approach multi-site brain MRI landmark detection pipeline predicting 51 anatomical landmarks across six simulated acquisition sites (642 labeled, 162 unlabeled subjects). A 3D CNN achieves the best internal accuracy at **4.54 mm** mean Euclidean error on a 20% per-site holdout and is submitted for all 162 unlabeled subjects. Key findings: rigid ACPC-space registration is algebraically equivalent to the mean baseline; posterior-weighted centroids add noise for sub-structure boundaries; coordinate normalisation is critical for regression with targets far from the origin; orbital landmarks (LE, RE) remain the dominant error source at 11–13 mm across all approaches.

1 Site Characteristics

Six sites provide 107–108 labeled and 27 unlabeled subjects each (siteB: 106 labeled). Per subject: T1 NIfTI, T2, six tissue posterior maps (WM, GM, CSF, Ventricle Bodies, Globus Pallidus, Background); labeled subjects also have 51 ground-truth landmarks in physical space (mm).

Orientation. Calling `SimpleITK.GetDirection()` on all 642 T1 volumes returns the identity matrix, confirming images are axis-aligned in SimpleITK's internal LPS convention — consistent with what the EDA HTML reports show. There are no rotational mismatches between or within sites; the physical-to-voxel mapping reduces to translate-and-scale. (The FCSV landmark files use RAS convention, so x and y sign-flips are required when calling `TransformPoint`; see Section 3.)

Spacing. Constant within each site (std = 0); ranges $\approx$ 0.92–1.10 mm across sites. See Appendix Appendix A for per-site values.

Intensity scale. Raw T1 (and T2) intensity ranges differ dramatically across sites, as confirmed by the EDA HTML reports: siteA $[0, 4096]$ (12-bit unsigned); siteB $[0, 255]$ (8-bit); siteC $[0, 6000]$; siteD $[1500, 13200]$ (non-zero background floor — the scanner clips values below 1500); siteE $[0, 1]$ (already normalised to a unit scale); siteF $[-1000, 1000]$ (Hounsfield-unit-like convention, centred at zero). Within each site the range is constant across all subjects (zero intra-site variation in scale). A model trained on raw intensities from one site would fail on another: the pipeline applies z-score normalisation (`normalize_intensity`) before any resampling or feature extraction, mapping each volume to zero mean and unit variance and removing this inter-site disparity.

Physical origin and AC position. AC is always at $(0, 0, 0)$ mm for siteA (pre-ACPC-aligned), but varies by $\sigma \approx 17$ mm per axis for sites B–F (native scanner space, purely translational scatter — no rotation). The principal challenge of cross-site generalisation is therefore translational offset and mild spacing variation, not arbitrary orientation. The AC–PC distance is 27 mm ($\sigma \approx 1$ mm) across all sites.

2 Design Decisions

All approaches use a deterministic 80/20 per-site labeled split ($\approx$21 eval subjects per site) and report mean Euclidean error over all 51 landmarks. Table 1 summarises results; full per-site breakdowns are in Appendix Appendix B.

Approach 1 — Per-site mean in ACPC space (baseline, 4.59 mm). Each subject's landmarks

Table 1: Mean Euclidean landmark error (mm) on the 20% per-site held-out evaluation set. CNN (Approach 4) is selected for final submission.

Approach	**siteA**	**siteB**	**siteC**	**siteD**	**siteE**	**siteF**	**Overall**	
1. Per-site mean (ACPC)	4.86	4.34	4.61	5.03	4.13	4.55	4.59	
2. Rigid registration	4.86	4.34	4.61	5.03	4.13	4.55	4.59	
3. Posterior heuristic*	5.13	—	—	—	—	—	5.13	
4. 3D CNN regression	4.54	4.38	4.64	5.04	4.13	4.53	**4.54**	✓
5. Multi-atlas affine	5.91	5.47	5.56	5.77	5.09	5.27	5.51	
6. BCD-inspired LLS	4.38	4.96	5.08	5.68	4.93	4.82	4.97	

*Applied to siteA only.

are transformed to ACPC space, averaged, then mapped back to scanner space via the target subject's ACPC transform. Averaging must occur in ACPC space — scanner-space averaging would fail for sites B–F due to the $\sigma \approx 17$ mm AC scatter. RE (12.4 mm) and LE (11.4 mm) dominate error; AC and PC are near-perfect by construction.

Approach 2 — Rigid registration to ACPC template (null result, 4.59 mm). A mean T1 ACPC template is built per site; unlabeled subjects' T1s are rigidly registered (6 DOF, Mattes MI) to propagate the mean landmark set. This matches Approach 1 exactly: the registration refines the ACPC transform estimate, but the propagated landmarks remain the population mean. Rigid registration of a mean cannot reduce anatomical variation around that mean; improvement requires subject-specific image-content learning. The per-subject approximate ACPC transform T_{approx} produced here is reused by Approaches 3, 4, and 6.

Approach 3 — Posterior-guided local refinement (negative result, 5.13 mm, siteA only). For eight CC sub-structure landmarks, the mean prediction was replaced with a WM-posterior-weighted centroid within an 8 mm sphere. Score regressed by +0.54 mm (genu +4.42 mm, rostrum +4.62 mm) because P_{WM} is high throughout all white matter, not specifically at CC boundaries. Posterior-weighted centroids are only valid when the landmark coincides with the tissue centre of mass (e.g., ventricular body).

Approach 4 — 3D CNN coordinate regression (best, 4.54 mm, submitted). A 5-block stride-2 encoder CNN (1→24→48→96→192→384 channels) followed by global average pooling and two FC layers (384→256→153, Tanh output) predicts all 51 landmarks from the ACPC-resampled T1 (96×101×101, 2 mm). Trained cross-site (≈516 subjects), 100 epochs, AdamW + cosine LR, smooth-L1 loss. *Critical fix:* the first run (no normalisation) yielded 14.1 mm; normalising targets by 120 mm to $[-1, +1]$ consistent with Tanh reduced this to **4.54 mm**. Without normalisation, a zero-initialised regression head predicts all landmarks near the origin — correct for AC but catastrophic for peripheral landmarks at ±100 mm. Cross-site pooling is natural because ACPC space places all subjects in the same frame.

Approach 5 — Multi-atlas affine registration (5.51 mm). Five siteA atlases registered (12 DOF, Mattes MI) to each target; landmarks transferred and fused by median per axis. Worse than baseline: affine registration in scanner space is destabilised by the B–F translational scatter, and imperfect registrations add noise to the median vote.

Approach 6 — BCD-inspired LLS from tissue posteriors (4.97 mm). Re-implements BRAINSConstellationDetector's [1] LLS step in Python. Six posteriors sampled at the 51 mean ACPC positions projected into scanner space yield a 306-dimensional feature vector; ridge regression ($\alpha = 10$) maps features to ACPC-space coordinates. Matches CNN on siteA (4.38 mm, where T_{approx}

is exact) but degrades for B–F (∼5 mm) where approximate ACPC transforms introduce systematic sampling errors.

3 Output Coordinate Space and the AC = (0,0,0) Pitfall

External evaluation compares predicted landmarks to withheld ground-truth in *scanner space* (native T1 coordinates). A naive implementation saves predictions in ACPC space, where AC is forced to $(0, 0, 0)$ mm by construction — correct for siteA (which is already ACPC-aligned) but wrong by ∼40 mm for sites B–F where the physical AC lies far from the origin.

Initial (incorrect) output. All 162 unlabeled predictions were saved in ACPC space. For siteA, errors were small (∼4.5 mm) because ACPC ≈ scanner space. For siteB–F, the AC error alone was ∼40 mm; overall errors exceeded 45 mm.

After the fix. Each unlabeled subject's predictions are converted to scanner space using T_{approx}^{-1} (the inverse of the approximate ACPC transform from Approach 2). A secondary bug was that `SimpleITK.TransformPoint` operates in LPS convention while FCSV files use RAS, requiring sign-flips on x and y before and after the call. After both fixes, siteB unlabeled AC reads $(-33.4, -6.3, -5.3)$ mm — consistent with the observed siteB AC scatter of $\sigma \approx 17$ mm. Estimated scanner-space error for unlabeled subjects is ≈9.3 mm overall (vs. 4.54 mm on labeled holdout where T_{true} is available).

4 Verification and Reproducibility

Internal evaluation uses a deterministic 20% per-site holdout. **Geometric verification** checks AC at $(0, 0, 0)$, $|\text{AC}_{SI} - \text{PC}_{SI}| < 0.1$ mm, $|\text{AC}_{LR} - \text{PC}_{LR}| < 0.1$ mm, and $|\text{LE}_{SI} - \text{RE}_{SI}| < 0.1$ mm for all labeled training subjects (HTML report: `eda_report_acpc_verification.html`). **Test suite:** 40 unit tests covering ACPC transform, error metrics, and FCSV I/O pass under `pixi run test`.

Reproducibility. A new user on the MIATT workstation in July 2026: clone the repository, run `pixi install` (≈5 min, reads `pixi.lock` — fully pinned), verify with `pixi run test`, then `pixi run pipeline --approach cnn --output predictions`. The CNN checkpoint, ACPC templates, and per-subject ACPC caches are generated on first run and reused thereafter; re-runs complete in under five minutes. No site-specific configuration is required.

5 Landmark Selection for Non-linear Registration

Mislocalised landmarks can invert a non-linear registration locally [6]. Based on CNN errors (Table 10, Appendix Appendix C): **exclude** all 15 with mean error > 5 mm (RE 12.6, LE 11.2, cortical poles/sulcal extremes 6–9 mm, lateral inner ear 7–8 mm, dens_axis 5.7 mm — orbital regions or cortical surfaces, low T1 contrast); **hard constraints** — the 26 with error ≤ 4 mm (AC, PC, corpus callosum family, caudate heads, lateral ventricles, optic chiasm, BPons, SMV, CM — distinct anatomy spanning all three axes); **soft constraints with reduced weight** — the 10 at 4–5 mm (l/r_corp, l/r_prim_ext, l_temp_pole, mid_prim_sup).

Conclusion. The 3D CNN achieves **4.54 mm** on the internal holdout and is submitted. LE and RE (11–13 mm) are the dominant bottleneck; adding T2 as a second input channel is the highest-leverage future improvement. The pipeline requires zero code changes for a new site by design.

References

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Appendix A EDA: Site Characteristics Tables

Table 2: Per-site dataset characteristics confirmed by EDA across all 642 labeled subjects.

Characteristic	**siteA**	**siteB**	**siteC**	**siteD**	**siteE**	**siteF**
Labeled subjects	108	106	107	107	107	107
Unlabeled subjects	27	27	27	27	27	27
Voxel spacing (mm, approx.)	1.00	1.02	1.01	1.03	0.98	1.03
Intra-site spacing std	0.000	0.000	0.000	0.000	0.000	0.000
Direction cosines	Identity in SimpleITK LPS — axis-aligned, no rotation					
AC scatter σ (mm/axis)	$\approx$0	$\approx$17	$\approx$17	$\approx$18	$\approx$17	$\approx$17
Pre-ACPC-aligned?	Yes	No	No	No	No	No
AC–PC distance (mm)	27.0 ± 1.0 (mean $\pm$ std)					
Notable artifacts	—	—	`.DS_Store`	—	—	—

Table 3: Image size range and physical FOV per site.

Site	**Size range (voxels, LR$\times$AP$\times$SI)**	**Spacing (mm)**	**Approx. FOV (mm)**
siteA	176–200 $\times$ 204–236 $\times$ 168–198	1.00	176–200
siteB	248–271 $\times$ 295–322 $\times$ 274–299	1.02	253–276
siteC	273–298 $\times$ 270–297 $\times$ 248–275	1.01	276–301
siteD	230–255 $\times$ 276–305 $\times$ 266–295	1.03	237–263
siteE	281–305 $\times$ 299–331 $\times$ 236–260	0.98	275–299
siteF	249–270 $\times$ 257–289 $\times$ 269–292	1.03	256–278

Appendix B Full Per-Approach Results

Table 4: Approach 1 — Per-site Mean in ACPC Space.

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.86	1.67	10.79
siteB	85	21	4.34	1.18	7.22
siteC	86	21	4.61	1.77	9.48
siteD	86	21	5.03	2.31	11.17
siteE	86	21	4.13	1.40	7.50
siteF	86	21	4.55	1.58	8.95
Overall			**4.59**	1.71	11.17

Table 5: Approach 2 — Rigid Registration to ACPC Template (null result).

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.86	1.67	10.79
siteB	85	21	4.34	1.18	7.22
siteC	86	21	4.61	1.77	9.48
siteD	86	21	5.03	2.31	11.17
siteE	86	21	4.13	1.40	7.50
siteF	86	21	4.55	1.58	8.95
Overall			**4.59**	1.71	11.17

Table 6: Approach 3 — Posterior-Guided Local Refinement (siteA only; negative result).

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	5.13	1.66	11.07
Overall			**5.13**	1.66	11.07

Table 7: Approach 4 — 3D CNN Direct Coordinate Regression (submitted).

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.54	1.79	10.92
siteB	85	21	4.38	1.14	6.57
siteC	86	21	4.64	1.83	9.56
siteD	86	21	5.04	2.61	11.79
siteE	86	21	4.13	1.17	7.36
siteF	86	21	4.53	1.71	9.88
Overall			**4.54**	1.80	11.79

Table 8: Approach 5 — Multi-Atlas Affine Registration.

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	5.91	2.34	13.19
siteB	85	21	5.47	1.27	8.36
siteC	86	21	5.56	2.19	11.02
siteD	86	21	5.77	2.44	12.23
siteE	86	21	5.09	1.73	8.83
siteF	86	21	5.27	1.81	10.77
Overall			**5.51**	2.02	13.19

Table 9: Approach 6 — BCD-Inspired LLS from Tissue Posteriors.

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.38	1.27	7.30
siteB	85	21	4.96	1.30	7.77
siteC	86	21	5.08	1.98	10.20
siteD	86	21	5.68	2.40	12.64
siteE	86	21	4.93	1.36	7.60
siteF	86	21	4.82	1.69	9.45
Overall			**4.97**	1.76	12.64

Appendix C Landmark Error Analysis (CNN, All Sites)

Table 10: Mean Euclidean error per landmark (Approach 4 — CNN), averaged across all six sites. Sorted lowest to highest. The row marked *(excluded)* separates the recommended ≤ 5 mm selection threshold from the excluded set.

Landmark	**Error (mm)**	**Landmark**	**Error (mm)**
AC	0.06	mid_prim_sup	4.24
PC	0.93	l_prim_ext	4.51
rostrum	1.55	r_corp	4.83
mid_lat	1.63	l_temp_pole	4.87
mid_basel	2.01	l_corp	5.14
RP	2.14	*(excluded, >5 mm)*	
genu	2.21	dens_axis	5.66
RP_front	2.38	r_lat_ext	5.85
lat_right	2.52	r_temp_pole	6.05
SMV	2.56	l_lat_ext	6.12
lat_left	2.58	l_occ_pole	6.21
rostrum_front	2.95	l_front_pole	6.50
CM	2.98	right_lat_inner_ear	6.63
l_inner_corpus	3.01	r_occ_pole	6.98
left_cereb	3.06	r_front_pole	7.64
l_caud_head	3.12	left_lat_inner_ear	7.71
r_inner_corpus	3.20	top_right	8.02
mid_sup	3.22	l_sup_ext	8.03
optic_chiasm	3.23	top_left	8.15
r_caud_head	3.30	r_sup_ext	8.22
lat_ven_left	3.39	LE	11.23
mid_prim_inf	3.42	RE	12.56
VN4	3.46		
m_ax_sup	3.66		
lat_ven_right	3.67		
callosum_right	3.72		
BPons	4.04		
m_ax_inf	4.12		
r_prim_ext	4.19		
callosum_left	4.20		

Appendix D CNN Architecture and Training Details

Table 11: 3D CNN encoder architecture (Approach 4).

Block	**In ch.**	**Out ch.**	**Spatial stride**	**Output size**
Input (ACPC T1)	1	—	—	$1 \times 96 \times 101 \times 101$
Conv block 1 (stride-2)	1	24	2	$24 \times 48 \times 50 \times 50$
Conv block 2 (stride-2)	24	48	2	$48 \times 24 \times 25 \times 25$
Conv block 3 (stride-2)	48	96	2	$96 \times 12 \times 12 \times 12$
Conv block 4 (stride-2)	96	192	2	$192 \times 6 \times 6 \times 6$
Conv block 5 (stride-2)	192	384	2	$384 \times 3 \times 3 \times 3$
Adaptive avg pool	384	—	—	$384 \times 1 \times 1 \times 1$
FC1 + ReLU	384	256	—	256
FC2 + Tanh	256	153	—	153

Each conv block = Conv3d(stride=2) → BN → ReLU → Conv3d(stride=1) → BN → ReLU. Output 153 values = 51 landmarks × 3 coordinates, normalised to $[-1, +1]$ by dividing by `COORD_SCALE`= 120 mm.

Table 12: Training hyperparameters.

Parameter	**Value**
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$)
Learning rate schedule	Cosine annealing, $\eta_{\max} = 10^{-3}$, $\eta_{\min} = 10^{-5}$
Epochs	100
Batch size	4
Loss function	Smooth L1 (Huber loss, $\delta = 1.0$)
Augmentation	Intensity jitter ±5%; left-right flip with $p = 0.5$
Training subjects	≈516 (80% of all 6 sites combined)
Coordinate scale	120 mm
GPU	NVIDIA RTX 4000 Ada (20 GB VRAM)
Training time	≈45 minutes